CAMPUS SKILLING, AUTOMATED EVALUATION & CORPORATE PLACEMENT SYSTEM

AptRun User Guide: How to Use Every Feature

A certified, step-by-step operational handbook for **Students**, **College TPOs**, **Faculty Section Coordinators**, and **Enterprise Recruiters**. Learn how to navigate every dashboard, launch containers, run proctored tests, track readiness, and manage campus hiring pipelines.

STATUTORY CORPORATE REGISTRY & INTELLECTUAL PROPERTY ATTESTATION MCA STATUTORY RECORD | LEGALLY VALIDATED

PLATFORM PRODUCT	PROPRIETARY BRAND	PARENT CORPORATE BODY	CORPORATE IDENTITY NO. (CIN)
AptRun™ Platform (aptrun.com)	Mellow Vault (mellowvault.com)	PRAYUKTI DEVELOPMENT PRIVATE LIMITED	U62099UP2024PTC207212
STATUTORY AUTHORITY	**REGISTERED OPERATIONS BASE**	**ENTERPRISE SUPPORT & DESK**	**INFORMATION SECURITY STANDARDS**
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Official Validity & Authenticity Notice: This document serves as the official, authenticated operational user handbook for the AptRun platform, engineered and owned by Mellow Vault, a unit of PRAYUKTI DEVELOPMENT PRIVATE LIMITED (CIN: U62099UP2024PTC207212). All workflows, student scoring algorithms (AptRun Readiness Score v3/v4), proctoring compliance protocols, tenant controls, and placement pipelines documented herein represent binding operational standards for participating educational institutions, candidates, and enterprise recruiters.

First-Time Onboarding: How Every User Begins

1 Student First Login & Track Selection
Navigate to `/login` or follow your college email link. The wizard prompts you to:
- **Confirm Branch:** Resolves to CSE, ECE, EEE, Mechanical, or Civil.
- **Select Track Focus:** Choose *IT First*, *Core First*, or *Balanced*.
- **Primary Language:** Pick C++, Java, Python, or "I don't code yet".

Where: /signup or First-Login Modal → Settings > Branch & Track

2 TPO / Coordinator Activation
Placement Officers and Faculty sign in with university credentials.
- **TPOs:** Land on `/admin` (TPO Command Center) with batch funnel metrics.
- **Coordinators:** Land on `/coordinator` with assigned section rosters.
- **Temporary Logins:** Must set a personalized password upon first entry.

Where: /login → Automatic role redirection to /admin or /coordinator

The 4 Primary Operational Personas in this Playbook

USER ROLE	DASHBOARD ROUTE	YOUR MAIN DAILY OBJECTIVE	KEY ACTION
Student Coder	`/dashboard` & `/arena`	Solve DSA, take proctored tests, build full-stack projects, raise ARS to ≥ 60, get hired.	DAILY PRACTICE & VAULT
College TPO	`/admin` (TPO View)	Onboard batch, track readiness, map campus drives, delegate faculty, export 7 reports.	COHORT & DRIVE FUNNEL
Faculty Coordinator	`/coordinator`	Check 12-student "Needs Attention" queue, log counseling calls, monitor live section tests.	REMEDIAL INTERVENTIONS
Corporate Recruiter	`/admin/company`	Propose campus drives, scout verified ARS talent, author coding tests, review AI mock videos.	DIRECT CAMPUS SOURCING

Navigation Quick Tip: The Real-Time Plan Signal
AptRun features an automatic real-time sync heartbeat (`/api/me/plan-signal`). When your college upgrades your plan or your subscription changes, sidebar locks and feature limits update within 30 seconds automatically—**no page refresh required**.

How to Solve Algorithmic Problems: Step-by-Step

The Practice Arena (`/dashboard/practice`) is your core developer training ground. Follow this workflow to solve problems efficiently:

1 Browse & Filter Problems

Open **Practice Arena** from the sidebar. Use the top filters:
- **Difficulty:** Filter by *Easy*, *Medium*, or *Hard*.
- **Topic Tags:** Select Arrays, Trees, Graphs, Dynamic Programming, Strings, Greedy.
- **Company Tags:** Filter problems asked at Google, Amazon, Microsoft, TCS, Infosys.
- A green checkmark appears on cards for problems you have already accepted (AC).

UI: Click problem title → Opens Monaco Editor View (/dashboard/practice/view/[id])

2 Select Your Programming Language

In the editor header dropdown, select from **19 supported programming languages**: C, C++, Java, Python 3, Go, Rust, TypeScript, JavaScript, C#, PHP, Kotlin, Swift, etc.
- The editor automatically loads language-specific starter boilerplate and main execution templates.
- *Auto-Save:* AptRun caches your unsubmitted code locally per language so you never lose drafts.

Shortcut: Use the language selector dropdown on top right of the editor toolbar

3 Run Code with Custom Test Inputs

Before submitting, verify edge cases using the test panel:
- Check the **"Custom Input"** checkbox below the editor.
- Enter your custom standard input (stdin) in the text area.
- Click `Run Code` (or press `Ctrl` + `'`).
- Inspect output: stdout, stderr, execution wall-clock time (ms), and peak memory consumed (KB).

Execution Speed: Sub-second execution (<850ms) powered by isolated Piston runners

4 Submit Code & Understand Verdicts

Click the green `Submit` button (or `Ctrl` + `Enter`). Your code is evaluated against hidden test cases.

Submission Verdict Reference Table

VERDICT	MEANING	WHY IT HAPPENS	WHAT YOU SHOULD DO
AC (ACCEPTED)	100% Correct	Your code passed all standard and hidden edge test cases within time and memory limits.	Problem marked solved; points added to your Readiness Score.
WA (WRONG ANSWER)	Incorrect Output	Logic failed on specific edge cases (e.g., negative numbers, empty arrays, large integers).	Check the failed test input displayed on screen and debug your logic.
TLE (TIME LIMIT)	Algorithm Too Slow	Time complexity exceeded (e.g. $O(N^2)$ instead of $O(N \log N)$ on $N=10^5$).	Optimize your algorithm; replace nested loops with hash maps or binary search.
MLE (MEMORY LIMIT)	Exceeded 256 MB	Infinite recursion or allocating excessive matrix memory in memory-constrained environment.	Check recursion base cases and optimize auxiliary space structures.
CE (COMPILE ERROR)	Syntax Error	Missing semicolons, type mismatches, or missing library imports.	Read the compiler output log displayed in the red console banner.

How to Inspect Past Submissions

Click the **Submissions** tab below the problem description. Click on any past attempt to open a modal displaying the exact code you submitted, the verdict received, execution time, and date stamp.

How to Take a Proctored Contest: Complete Walkthrough

AptRun hosts weekly rated programming contests and college mock assessments (`/dashboard/contests`). Follow these instructions to avoid disqualification:

1 **Register & Check Countdown**

Open **Contests** in the sidebar. Contests are grouped into *Upcoming*, *Live Now*, and *Past*.

• Click `Register` on upcoming contests. When the timer hits zero, click `Enter Contest`.

• *Audience check:* College mock contests are visible only if mapped to your university by your TPO.

UI: Click Enter Contest on the live contest card (/dashboard/contests/view/[id])

2 **Complete Pre-Flight Hardware Checks**

Before entering the problem console, the anti-cheat proctoring wizard runs hardware checks:

• **Camera Permission:** Allow browser webcam access. Position yourself centered and clearly lit.

• **Microphone Permission:** Verify audio levels to ensure background noise monitoring is active.

• **Browser Check:** Close secondary monitors, screen-sharing apps (Zoom, Discord), and browser devtools.

Warning: Denying camera permissions blocks entry into proctored tests

3 **Enter Forced Fullscreen Environment**

Click `Enter Fullscreen & Start Test`. The browser locks into full-screen mode.

• The test clock synchronizes with the server. A countdown appears at the top center.

• Use the top **Problem Palette** (Problem A, B, C, D) to navigate between challenges.

Rule: Do NOT press Esc or switch tabs during the contest

Understanding the Anti-Cheat 3-Strike Rule

Strike 1
FIRST WARNING
Tab blur or fullscreen exit. Screen shows yellow alert banner. Recorded in audit log.

Strike 2
FINAL WARNING
Second violation. Yellow-orange modal alerts you: "One more strike will terminate your test".

Strike 3
LOCKED OUT
Automatic submission of on-screen code. Access ends immediately. Only TPO can reinstate.

What Triggers an Anti-Cheat Strike?

ACTION	RESULT	WHY IT'S ENFORCED
Switching Browser Tabs	STRIKE ISSUED	Window blur event detected immediately by client and verified by proctoring server.
Exiting Fullscreen Mode	STRIKE ISSUED	Pressing `Esc` or minimizing window triggers immediate security alert.
Webcam Obstruction / Leaving Seat	FLAGGED IN LOG	AI detects absence of face or multiple faces in camera feed; flagged for TPO inspection.
Opening Inspect / DevTools	IMMEDIATE STRIKE	Keyboard shortcuts like `F12` or `Ctrl` + `Shift` + `I` are intercepted.

Understanding Your Skill Rating & Divisions

Rated contests update your **Elo Skill Rating** (starts at 1200). Divisions are: **Division 3** (<1400 rating), **Division 2** (1400–1799 rating), and **Division 1** (≥1800 rating). Perform well to earn 1 to 7 star badges displayed on your public Vault profile.

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How to Use Core Prep (ECE, EEE, Mechanical, Civil)

If you are an engineering student in a non-CS department, Core Prep (`/dashboard/core`) gives you specialized technical preparation. Here is how to use it for topic mastery and timed mock exams:

1 **Verify Your Branch & Track Focus**

In **Settings > Branch & Track**, ensure your branch is correctly mapped:

- **Electronics & Comm. (ECE):** Signals, Digital VLSI, Analog Circuits, Microprocessors.
- **Electrical Engineering (EEE):** Power Systems, Machines, Control Systems, Power Electronics.
- **Mechanical Engineering (ME):** Thermodynamics, SOM, Fluid Mechanics, Kinematics, CAD/CAM.
- **Civil Engineering (CE):** Structures, RCC & Steel, Geotechnical, Surveying, Transportation.
- Set Track Focus to **"Core First"** (places Core Prep right under Overview) or **"Balanced"**.

UI: Sidebar shows "Core Prep" with a "New" badge as soon as your branch is live

2 **Daily Topic Practice (Instant Worked Solutions)**

Click on **Topic Practice**:

- Select a subject module (e.g. *Power Electronics* or *Strength of Materials*).
- Select an individual topic → Start answering questions.
- **Instant Feedback:** After selecting an option, click `Check Answer` . AptRun immediately displays whether you are correct, accompanied by a full step-by-step worked derivation with KaTeX equations.

Limits: Free plan: 10 Qs/day. Expert / College plan: Unlimited practice

3 **Take Timed Proctored Core Tests**

Navigate to **Core Tests** tab (`/dashboard/core/tests`). Click `Start Test` on available papers:

- **Server-Enforced Clock:** Countdown timer runs at top right; test submits automatically when time expires.
- **Negative Marking:** Watch for penalty schemes (+1.0 for correct, -0.25 for wrong).
- **On-Screen Calculator:** Click the floating calculator icon to open a scientific calculator for complex calculations.
- **Question Palette:** Navigate questions, mark questions for review, and inspect answered/unanswered counts.

Where: /dashboard/core/session?test=[id]

How Core Mastery Feeds Your Readiness Score

CORE COMPONENT	WEIGHT IN CORE ARS	HOW IT IS CALCULATED & HOW TO MAXIMIZE IT
Timed Mock Tests	**60% of Core Score**	Average percentage scored across your last 5 completed timed Core tests within the last 180 days. Take mock tests regularly to keep recent form high.
Syllabus Topic Mastery	**40% of Core Score**	Percentage of your branch's topics where you have achieved **Mastered** status (≥5 answers with ≥75% accuracy over 180 days across practice and tests).

The 30-Day Go-Live Grace Period

When your college switches your branch live on Core Prep, a **30-day grace period** protects your score: the server computes your score under both the Software and Core formulas and automatically **gives you the higher score**, ensuring you never drop in rank while building your subject record.

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1. Vocabulary Sprint: Mastering Words with Spaced Repetition (SRS)

The Vocabulary module (`/dashboard/learning-centre/vocabulary`) trains you on 160 high-impact interview words using the Leitner 6-box model.

1 **Start Your Daily Sprint (3 Minutes)**
Open **Vocabulary Sprint** → Click `Start Daily Sprint` (~12 questions).
• The algorithm queues words due for review today, plus 2–4 new words from your active deck.
• **Meet the Word:** New words are introduced first with audio pronunciation, definition, example sentence, and usage tip.
Goal: Complete 10 answers daily to maintain your practice streak

2 **Answer Progressive Question Types**
As words advance across Leitner boxes, questions increase in difficulty:
• *Level 1:* Definition match → *Level 2:* Pick word for a meaning → *Level 3:* Sentence blank → *Level 4:* **Type from memory** (spelling counts; single-letter slips are flagged as near misses).
• Correct answers advance words toward Box 5 (Mastered). Mistakes drop words 2 boxes and schedule review today.

2. Audio-First Speaking Practice with Pronunciation Feedback

Speaking Practice (`/dashboard/learning-centre/speaking`) coaches your spoken English clarity, fluency, and articulation without human bias.

1 **Choose a Reading Passage or Generate a Custom AI Topic**
Browse the passage library (*Interview Ready*, *Workplace English*, *Tech & Career*, *Everyday English*), or click `AI Custom Passage` and enter any topic (e.g. "My React project", "How DNS works").
UI: Click Start Practice on any passage card

2 **Record Your Spoken Answer Aloud**
Click `Start Recording`. A live audio waveform confirms microphone capture.
• Read the passage naturally. Avoid rushing or whispering. Click `Stop & Submit` when finished.
• *Hardware voice detector:* Silent or muted recordings are rejected locally to preserve requests.

3 **Review Word-by-Word Alignment & Pronunciation Notes**
AptRun analyzes your raw audio using `SpeechAlignment` and Gemini speech models:
• **Overall Score (0-100):** 0.35 × Accuracy + 0.35 × Clarity + 0.30 × Fluency. Pass mark is ≥ 60%.
• **Word Heatmap:** Words highlighted green (accurate), yellow (hesitation), or red (mispronounced/omitted).
• **"Worth a Second Listen":** Specific words flagged for acoustic coaching.
Privacy: Audio recordings are processed in memory and never permanently stored

3. Listening Lab: Corporate Dialogue Comprehension

How to Complete a Listening Lesson (`/dashboard/learning-centre/listening`)
Select from 24 graded audio lessons. Click `Play Audio` to listen to simulated corporate client briefings or standup calls. Answer the comprehension questions before the timer expires. Your score feeds the Learning Centre component of your ARS.

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1. Soft Skills Tests: How to Practice the 70 Exam Sets

The Soft Skills portal (`/dashboard/soft-skills`) contains 70 pre-calibrated examination papers calibrated to corporate placement exams.

1 **Select an Exam Set or Assigned Test**

Open **Soft Skills**. The page displays tests in two sections:

- **Assigned to You:** Mandatory tests scheduled by your College TPO or a visiting hiring partner.
- **AptRun Question Sets (Sets 1 to 70):** Open library. Search by number (e.g. "Set 14") or filter by status: *Not Attempted*, *In Progress*, or *Completed*.

UI: Click Start Test or Resume Test on any test card

2 **Complete the 25-Question Exam**

Each set contains 25 questions with a 30-minute timer:

- **8 Quantitative Aptitude · 7 Logical Reasoning · 6 Verbal Ability · 4 Situational Judgement.**
- Navigate using the bottom question dots. Click `Submit Test` when finished.
- *Instant Results:* View your overall score, category percentiles, and full answer review with explanations.

2. Project Lab: Building Real-World Apps in In-Browser VS Code

Project Lab (`/dashboard/projects`) provides containerized cloud developer workspaces to build and deploy portfolio-grade applications.

1 **Browse Projects & Download Client Statement of Work (SOW)**

Browse projects across categories: E-Commerce, Healthcare, EdTech, FinTech.

- Click on a project card → Click `Download SOW PDF`.
- Read the client brief, business requirements, API specifications, and milestone deliverables.

Entitlement: Included in Expert and Grandmaster plans (80 workspace-hours/month)

2 **Launch Your Dedicated Cloud Workspace (1-Click)**

Click `Launch Workspace`. AptRun spins up an isolated OpenVSCode Server container:

- **Left Panel:** Full-featured VS Code in browser with terminal, file tree, extensions, and git.
- **Right Panel:** Live web preview rendering your running application via reverse Caddy proxy.
- *Auto-Reap:* Workspaces idle for 15 minutes stop automatically to conserve your monthly hours.

Shortcut: Open integrated terminal with Ctrl + ` to run npm install or npm run dev

3 **Submit Milestones & Request Review**

Tick off completed milestones in the workspace sidebar checklist. When finished:

- Click `Submit Code for Review`. The system snapshots your container filesystem and triggers automated verification.
- Verified projects earn an authenticated badge in your Mellow Vault portfolio.

Storage & Volume Persistence

Your code is saved on a persistent volume (`wsvol-<id>`). When your workspace stops, your files remain intact. Relaunching the workspace restores your exact directory state.

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1. AI Mock Interviews: Speech-to-Speech Practice

The AI Interview portal (`/dashboard/interviews`) simulates technical and HR panels using speech-to-speech AI.

1 **Select an Interview Track or Role Template**

Click `New Mock Interview` . Choose a role template (e.g. *Full Stack Engineer*, *Data Analyst*, *Embedded Systems*), or select a company track (Google, Amazon, TCS Digital).

UI: Click Start Session → Allow microphone permission

2 **Conduct the Interview Session**

The synthetic AI interviewer asks a verbal question. Listen through your speakers/headphones.

- Click `Speak` and answer verbally (or use the text input toggle).
- The AI listens, evaluates your answer, and asks dynamic follow-up questions to probe depth.
- Click `Conclude Interview` when all rounds finish.

3 **Inspect Your Rubric Scorecard & Transcript**

Review your post-interview report:

- **4 Rubric Scores:** Technical Correctness (35%), Problem Solving (30%), Communication (20%), Code Quality (15%).
- **Verbatim Transcript:** Read the full conversation with inline technical feedback.
- Your last 5 scored interviews directly drive 20% of your Placement Readiness Score.

2. How to Raise Your Placement Readiness Score (ARS v4)

+20 Pts	+15 Pts	+15 Pts	+15 Pts
AI INTERVIEWS	CONSISTENCY	CODING BREADTH	LEARNING CENTRE
Maintain ≥80% avg across last 5 interviews	Practice 12 days in the trailing 28 days	Solve 20 Easy, 40 Med, 40 Hard problems	Complete daily vocabulary sprints and speaking

3. How to Enter The Vault & Apply to Campus Drives

1 **Unlock Vault Eligibility (ARS ≥ 60)**

When your ARS score crosses **60 points** and you hold an eligible plan (Expert, Grandmaster, or College), open **Vault** (`/dashboard/vault`).

- Toggle the **"List My Profile in Mellow Vault"** switch to ON.
- Your verified badges, project repos, and contest stats become visible to hundreds of corporate recruiters.

Where: /dashboard/vault → 1-click opt-in toggle (revocable anytime)

2 **Apply to Campus Placement Drives**

Scroll to **Upcoming & Drives** on your dashboard:

- The system checks your CGPA, backlogs, branch, and ARS score against company criteria.
- Click `Apply Now` on eligible drives. Track your stage live: `Applied → Shortlisted → OA → Interview → Offer` .

How to Onboard Your Entire Student Batch: Step-by-Step

The College TPO portal (`/admin`) enables placement teams to bring thousands of students onto AptRun in minutes:

1 **Prepare the Student Excel / CSV Roster**
Format your student data sheet with the standard 9 columns:
`Name, Roll_Number, Email, Phone, Parent_Phone, Branch, Section, CGPA, Backlogs`
• *Branch normalization:* Variations like "MECH", "ME", and "Mechanical" are automatically mapped.
• *Section names:* Whitespace and casing are normalized ("A", "a ", "A " are treated as Section A).
UI: /admin/students → Click "Import Students" button on top right

2 **Upload Roster & Review Dry-Run Validation**
Drag and drop your XLSX/CSV file. AptRun performs instant validation:
• Highlights duplicate roll numbers or existing emails.
• Shows how many new accounts will be created and how many existing records will be updated.
• Click `Confirm Import`. The server generates student accounts and sends welcome activation links.

How to Use the 29-Column Student Cohort Directory

Navigate to **Student Cohort** (`/admin/students`) to search and filter your entire student body:

Smart Filters Toolbar
Filter by **Branch**, **Section**, **Readiness Tier** (Ready, In Progress, Needs Training), **At-Risk (<60 ARS)**, **Drive Eligible Only**, or **Coding Language**.

Batch Multi-Select Actions
Select multiple student checkboxes to dispatch bulk email notifications, assign custom mock tests, or export filtered rosters to Excel.

Live Practice Streaks
Table columns display 7-day practice score, active streak, last active date, and current placement drive status in real time.

Using the Student 360° Management Drawer

Click on any student row in the cohort table to slide open their comprehensive 4-tab 360° drawer:

DRAWER TAB	INFORMATION DISPLAYED	TPO CONTROL ACTIONS
1. Overview	Roll Number, Personal Phone, Parent Phone, Branch, Section, Academic CGPA, Backlogs, Drive Eligibility status.	Edit contact information, correct CGPA/backlog errors, or re-map branch.
2. Placement Readiness	Full ARS v4 breakdown: Interview score, Coding score, Core Technical score, Consistency score, Soft Skills score.	Click `Recalculate Readiness` or `Download Student PDF Dossier`.
3. Submissions	Chronological log of the student's latest 80 practice and contest attempts. Syntax-highlighted code viewer.	Inspect code quality, verify originality, and investigate plagiarism flags.
4. Coordinator History	Chronological log of all interventions logged by faculty: phone calls, meetings, notes, and disciplinary blocks.	Review faculty follow-up compliance; lift coordinator account holds.

How to Delegate Section Coordinators

1 **Assign Faculty to Sections & Branches**
Open **Section Coordinators** (`/admin/coordinators`) → Click `Add Coordinator`.
• Select faculty member email, designated Section (e.g. "Section A"), and Branch Scope (All or Specific branches).
• *Collision Check:* Server prevents two coordinators from covering the same branch in a section.
• Check the **"N not covered"** chip to find and assign unmapped student sections.

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1. How to Schedule Institutional Mock Assessments

TPOs can deploy mock coding contests, AI interview drives, and soft skill tests for their students:

1 **Schedule a Mock Coding Contest**

Open **Mock Contests** (`/admin/mock-contests`) → Click `Create Mock Contest`.
- Set Title, Start Time, Duration (e.g. 90 mins), and targeted branches/sections.
- Pick problems from AptRun's problem bank or clone standard Mellow contest papers.
- Toggle **Strict Proctoring** to ON (enforces webcam, fullscreen lock, and 3-strike rule).

UI: Click Publish → Test automatically appears on assigned students' dashboards

2 **Deploy Soft Skills Question Papers**

Open **Soft Skills** (`/admin/mock-soft-skills`) → Click `New Soft Skills Test`.
- Step 1: Enter Title, Duration (e.g. 30 mins), and Passing Percentage.
- Step 2: Click `Add Ready-Made Question Set` → Pick from Sets 1 to 70.
- Click `Add Set & Publish`. Questions are copied into your private test instance.

2. Live Anti-Cheat Proctoring Radar & Reinstatement

During live examinations, monitor real-time integrity via the Proctoring Console (`/admin/proctoring`):

Live Radar Grid: Displays active test-takers with webcam indicators, tab-switch blur counts, and strike tallies.
Incident Stream: Real-time alerts when students trigger Strike 1, Strike 2, or Strike 3 (Lockout).
How to Reinstate a Locked Student: If a student is locked out due to verified hardware glitch, click the student's row in the proctoring feed → Click `Reinstate Student` → Enter mandatory reason. The student can immediately resume the test without resetting previous answers.

3. How to Manage Campus Placement Drives

1 **Map Catalog Drives or Create Private Drives**

Open **Campus Drives** (`/admin/drives`):
- **Available to Map:** Browse corporate drives syndicated by AptRun hiring partners. Click `Map to College` to make the drive visible to your eligible students.
- **Create Private Drive:** Click `New Placement Drive` for visiting companies.

2 **Set Strict Eligibility Cutoffs & Track Funnel**

Configure eligibility rules: Minimum CGPA (e.g. 7.0), Max Backlogs (0), Allowed Branches, and Minimum ARS score (e.g. 65).
- Track applicants across stages: `Registered → Shortlisted → OA → Interview → Offered → Accepted`.
- Record final CTC packages (LPA) for accepted offers to feed institutional placement rate metrics.

4. How to Generate the 7 Institutional Placement Reports

One-Click Excel & PDF Report Generation (`/admin/reports`)

Select your target branch or section → Click `Download Report` on any of the 7 ready-made reports:
1. Student Cohort Master (XLSX) · **2. Placement Readiness Summary (XLSX)** · **3. At-Risk Student Roster (XLSX)** · **4. Campus Drive Shortlist (XLSX)** · **5. Assessment Scores (XLSX)** · **6. Language Competency (XLSX)** · **7. Section Performance (PDF)**.

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How Section Coordinators Manage Their Classroom: Daily SOP

As a Faculty Section Coordinator (`/coordinator`), your mission is daily student engagement, attendance tracking, and early remedial intervention before placement season begins.

1. Your Morning Routine: The 12-Student "Needs Attention" Queue

Every morning, open `/coordinator`. At the top of your dashboard sits the auto-ranked **Needs Attention** queue:

STUDENT STATUS	RISK POINTS	WHY THEY ARE IN YOUR QUEUE	YOUR ACTION
Locked out of Test	+50 PTS	Received 3 proctoring strikes during a test; currently locked out of assessments.	Investigate incident in proctoring log; notify TPO if reinstatement is justified.
Follow-Up Overdue	+45 PTS	A scheduled counseling call or remedial check-in date passed without completion.	Call student immediately; record notes in follow-up log.
Follow-Up Due Today	+35 PTS	A scheduled follow-up action is due today.	Conduct check-in meeting; mark task as Completed.
Never Signed In	+30 PTS	Account provisioned by TPO but student has never set their password.	Resend activation credentials or remind student in class.
Dormant Account	+25 PTS	Zero practice problems or tests completed in the last 14 consecutive days.	Message student to restart practice habit.
Needs Training Tier	+15 PTS	Readiness Score (ARS) is below 45 points (<60 is classified At-Risk).	Assign targeted topic practice or soft skills sets.

2. How to Log Follow-Up Interventions

1 **Select Student & Open Follow-Up Log**

In your section roster, click on the student row → Click `Add Follow-Up Note`.

- **Interaction Type:** Select *Phone Call*, *In-Person Meeting*, *Message*, or *Internal Note*.
- **Notes:** Enter details (e.g., "Student struggling with Graph algorithms; advised to complete Easy BFS sheet").
- **Follow-Up Due Date:** Set a due date (e.g. 7 days from now) to automatically schedule the next check-in.

Transparency: All coordinator notes automatically surface in the TPO master cohort drawer

3. Live Section Proctoring & Account Holds

Section-Level Live Proctoring (`/coordinator/proctoring`)

During college-wide mock tests, coordinators open their dedicated proctoring tab. This feed isolates **only students in your assigned classroom**, letting you spot absent students or webcam violations without sifting through the whole batch.

Placing Disciplinary Account Blocks

If a student commits repeated infractions, coordinators can place a temporary hold. Click `Block Account` on the student row. **A mandatory reason string is required.** Note: Coordinators can only lift blocks placed by themselves; TPO blocks must be lifted by TPO.

4. Exporting Section Reports

Navigate to **Section Reports** (`/coordinator/reports`). Download weekly section attendance spreadsheets and ARS growth metrics to present at department faculty meetings.

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How Corporate Recruiters Source & Hire on AptRun: Step-by-Step

The Hiring Partner portal (`/admin/company`) streamlines university recruitment, eliminating unverified resumes and manual screening:

1 **Discover Partner Colleges & Propose Campus Drives**

Open **Partner Colleges** (`/admin/company/colleges`):

- Browse accredited engineering colleges. Filter by Tier (Tier 1 / 2 / 3), State, and graduating student volume.
- Click `Propose Campus Drive` → Select your Job Opening (CTC package, dates, roles, cutoffs).
- The proposal lands directly in the College TPO's inbox. Upon approval, the drive goes live to eligible students.

UI: Track proposal statuses (Pending, Approved, Declined) directly in the college directory

2 **Post Job Openings & Configure the Hiring Funnel**

Open **Job Openings** (`/admin/company/drives`) → Click `Post New Opening` :

- **Role Details:** Title, Role Type (Full-time, Internship), Location, CTC Range (LPA).
- **Eligibility Criteria:** Set minimum CGPA, maximum allowed backlogs, and minimum ARS score threshold.
- **Selection Stages:** Configure your funnel: `Online Assessment → AI Mock → Tech Panel → HR Offer` .

3 **Scout Pre-Assessed Candidates in the Talent Pool**

Rather than waiting for applications, actively headhunt candidates in **Candidates > Talent Pool**:

- **Filter by Verified ARS:** Select candidates with ARS ≥ 75 (Placement Ready).
- **Filter by Language & Stack:** Find developers verified in C++, Java, Python, React, Go, etc.
- **Filter by Discipline:** Filter software vs core engineering branches (ECE, EEE, Mechanical, Civil).
- Click `Invite to Apply` or `Direct Interview Invite` to reach top candidates directly.

Guarantee: Zero unverified claims; all candidates in the Vault are actively tested

4. Managing the Candidate Hiring Pipeline

PIPELINE STAGE	CANDIDATE ACTIVITY & STATUS	RECRUITER ACTION
1. Applied	Candidate applied to drive or accepted invitation; academic eligibility pre-verified.	Review profile; 1-click batch invite to Assessment.
2. Assessment	Candidate completes proctored coding and aptitude test. Automated test scores logged.	Inspect code playback, anti-cheat flags, and ranking.
3. AI Interview	Candidate completes asynchronous voice AI interview round. Transcripts available.	Listen to recordings; review rubric scores.
4. Technical Panel	Candidate scheduled for live video interview with engineering leads.	Record interviewer feedback and recommendation.
5. Offer Extended	Official offer letter dispatched through platform with CTC details.	Track candidate offer acceptance in real time.

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1. How to Author Custom Technical Assessments

Hiring partners can author bespoke coding and soft skill tests (`/admin/company/assessments`) tailored to their tech stacks:

1 **Create Test & Select Problem Bank Questions**
Click `Create Assessment` → Map to your active Job Opening.
• Choose problems from AptRun's shared library or author custom proprietary coding challenges.
• Configure allowed languages (e.g. restrict to Java and C++ for backend roles).
• Add an aptitude screening section by attaching ready-made Soft Skills sets (Sets 1 to 70).
UI: Click Publish → Generate secure candidate invite links or dispatch via email

2 **Inspect Anti-Cheat Reports & Keystroke Playback**
Open **Proctoring** (`/admin/company/proctoring`) to review test integrity:
• Review candidate webcam snapshots taken throughout the assessment.
• Inspect window blur and fullscreen violation counts.
• **Code Replay:** Click `Replay Code` to watch a keystroke-by-keystroke playback of how the candidate developed their solution, detecting sudden pastes of pre-written code.

2. How to Review Asynchronous AI Mock Interviews

Recruiters evaluate candidate voice interview recordings in `/admin/company/interviews` :

Audio Playback with Transcript Sync
Listen to spoken candidate answers at 1.0x, 1.25x, or 1.5x speed. The verbatim transcript highlights words in sync with the audio, allowing you to evaluate verbal confidence, clarity, and technical terminology.

AI Rubrics with Human Score Overrides
Review Gemini's automated breakdown across Technical Correctness, Communication, and Problem Solving. Recruiters can adjust scores manually and append private interviewer notes before deciding on next rounds.

3. How to Verify Credentials in The Mellow Vault

The Vault Verification portal (`/admin/company/vault`) lets recruiters verify proof-of-work with zero reliance on unverified resumes:

CREDENTIAL DIMENSION	WHAT YOU CAN AUTHENTICATE	RECRUITER BENEFIT
Project Lab Code Repos	Inspect the actual source code, Git commit history, and live container deployment of the student's capstone project.	Confirms true software craftsmanship versus copied boilerplate tutorials.
Contest Elo Rating	Inspect verified performance curves across weekly timed proctored coding contests under 3-strike anti-cheat rules.	Proves genuine algorithmic speed and problem-solving under strict exam conditions.
Spoken English & Fluency	Review objective pronunciation accuracy scores (SpeechAlignment), WPM pacing, and situational judgment scores.	Accelerates screening for multinational client-facing roles.

4. Exporting Candidate Hiring Reports

Open **Hiring Reports** (`/admin/company/reports`). View candidate conversion funnels, offer-acceptance ratios, and download complete candidate spreadsheets formatted for corporate ATS import (Workday, Greenhouse, Lever).

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How to Operate the Superadmin Master Console

The Superadmin Master Control (`/superadmin`) is restricted to `role: superadmin`. It controls platform economics, university tenant provisioning, payment reconciliation, and infrastructure scaling.

1. Plans & Dynamic Pricing Management

1 **Update Plan Pricing in Real Time**
Open **Plans & Pricing** (`/superadmin?tab=plans`):
• Edit the **List Price** and **Sale Price** for the Expert plan (e.g. list $14,999, sale $12,999/yr).
• Changes apply immediately on the public pricing page and checkout without redeploying code.
• *Grandmaster Plan:* Marked `contact_only = true` . Shows "Sold by our team" and cannot be bought online.
UI: Click Save Changes → Pricing cache clears instantly

2 **Adjust Daily Feature Limits**
Click `Edit Limits` on any plan card:
• Modify daily coding limits, daily AI mock interview quotas, and monthly Project Lab workspace-hours (default: 80 hrs).
• Toggle access switches: `hackathon_access` , `vault_access` , `project_lab_access` .

2. Manual Subscription Overrides & Payment Grants

Superadmins can grant, extend, or terminate plans manually (`/superadmin?tab=subscriptions`):

Give a Plan: Click `Give Plan` → Pick student email → Select plan (Expert or Grandmaster) → Set term (1 to 1,825 days) → **Enter mandatory audit reason**. Choose *Free of Charge* or *Paid Outside Gateway* (records offline fee).
Extend Days: Adds days to `current_period_end` without altering historical pricing.
Cancel Plan: Reverts student immediately to free Coder tier. Cancelling a university or corporate plan requires re-entering the Superadmin master password.

3. Safe Impersonation ("View As User" Mode)

Write-Locked Diagnostic Impersonation
To troubleshoot user issues, open **Accounts & Logins** → Click `Manage` on any user → Click `View as this user` .
• Generates an impersonation token (`impersonation:<id>`).
• **Write-Lock Guarantee:** The middleware (`RejectWritesWhileImpersonating`) automatically blocks all state-mutating requests (POST, PUT, DELETE) with `403 Forbidden` , preventing accidental data corruption while inspecting a student or TPO dashboard.

4. Immutable Audit Ledger & Judge Scaling Infrastructure

Immutable Audit Log (`/superadmin?tab=audit`)
Every administrative modification records Actor ID, Target Model, Action verb, before/after JSON diff, user IP address, and mandatory reason. Guarantees complete compliance and accountability.

Judge Infrastructure (`/superadmin?tab=infrastructure`)
Monitor Piston runner worker pools, Redis execution queues, and WebSocket signaling. Benchmarked to sustain concurrent examination loads exceeding 1,000+ active students simultaneously.

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Role-by-Role Daily Action Checklist

USER ROLE	DAILY OPERATIONS CHECKLIST	PRIMARY TARGET METRIC
Student Coder	1. Solve 1-2 DSA problems · 2. Complete daily 3-min Vocabulary Sprint · 3. Take 1 Speaking/Interview mock · 4. Check active drives.	Maintain streak; push ARS ≥ 60 to unlock The Vault.
College TPO	1. Review batch placement funnel · 2. Inspect At-Risk (<60 ARS) students · 3. Map new campus drives · 4. Export weekly XLSX reports.	Achieve 100% batch drive eligibility and placement conversion.
Section Coordinator	1. Check top 12 "Needs Attention" queue · 2. Contact overdue students · 3. Log intervention notes · 4. Monitor live section mock tests.	Clear daily overdue follow-up tickets; lift unjustified blocks.
Corporate Recruiter	1. Check drive applications · 2. Search pre-assessed Vault talent · 3. Review AI mock interview audio/transcripts · 4. Issue offer letters.	Accelerate Time-to-Hire with pre-vetted candidates.
Superadmin	1. Check Piston Judge latency · 2. Review payment reconciliation logs · 3. Audit manual grants · 4. Monitor university onboarding leads.	Maintain 99.99% judge uptime and zero subscription leakage.

Troubleshooting & Frequently Asked Questions (FAQ)

Q: My webcam is blocked or test says "Hardware Not Found"
Fix: Click the lock/tune icon on your browser address bar → Set Camera and Microphone to *Allow*. Close any apps that use the webcam (Zoom, Teams, Discord) and refresh the page.

Q: I got 3 strikes and got locked out of a contest
Fix: Contact your College TPO or Section Coordinator immediately. They can open the proctoring incident log and click `Reinstate Student` to unlock your session without losing saved answers.

Q: My Project Lab workspace shows "Disconnected"
Fix: If you are idle for 15 minutes, your container stops automatically to save monthly hours. Simply click `Restart Workspace`. All your code on the persistent volume is restored.

Q: I bought Expert or got a grant but features still show locked
Fix: The client polls `/api/me/plan-signal` every 30s. If you don't see it immediately, navigate to **Plans & Billing** (`/dashboard/billing`) to trigger an instant signature refetch.

System Support & Escalation Directory

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Official Support & Helpdesk: support@mellowvault.com · Helpline: +91 76785 78295 · Registered Office: Noida / New Delhi, India

SYSTEM OPERATIONAL · 99.99%